DEPARTMENT OF COMPUTER SCIENCE
RAJAGIRI COLLEGE OF SOCIAL SCIENCES
(Autonomous)
KALAMASSERY - KOCHI - 683104



MASTER OF COMPUTER APPLICATIONS

DBMS LAB RECORD

NAME : MERIN JAI

SEMESTER : 1

REGISTER NO. : _____

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MASTER OF COMPUTER APPLICATIONS

CERTIFICATE

NAME : MERIN JAI

SEMESTER : 1

REGISTER NO. : _____

Certified that this is a bonafide record of work done by the student in the Software Laboratory of Rajagiri Department of Computer Science, Kalamassery.

Faculty in Charge

Dean, Computer Science

Internal Examiner

External Examiner

Place : Kalamassery
Date :

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Activity # 1
ER Diagram & Table Design

Description :

- E-R Diagram and table reduction
- Table descriptions

Date : 09/08/2023

ER Diagram & Table Design

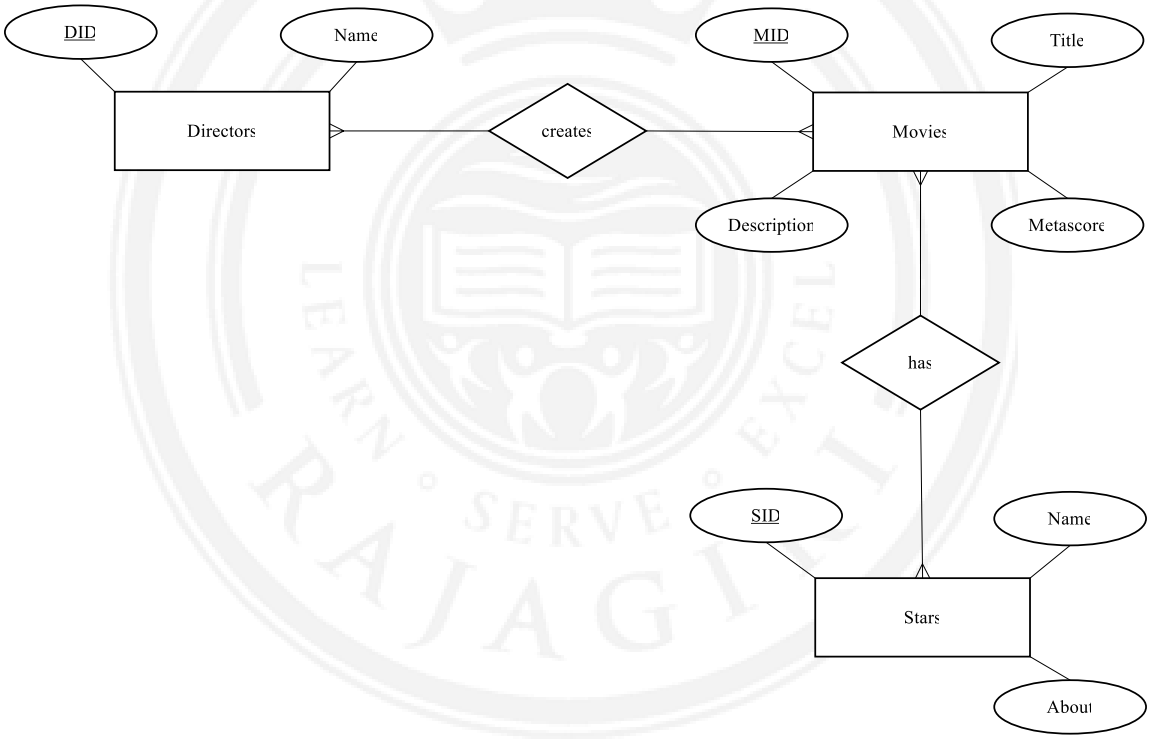


Table name: Directors
Description: Used to store Directors Information

Attribute	Data Type	Constraints
DId	Int	Primary Key/ Not Null
Name	Varchar2(40)	Not Null

Table name: Stars
Description: Used to store Stars Information

Attribute	Data Type	Constraints
SId	Int	Primary Key/ Not Null
Name	Varchar2(40)	Unique
About	Varchar2(100)	

Table name: Movies
Description: Used to store Movies Information

Attribute	Data Type	Constraints
MId	Int	Primary Key/ Not Null
Title	Varchar2(40)	Not Null
R_date	Date	
Image_url	Varchar2(100)	
Certificate	Varchar2(20)	
Runtime	Number(3,2)	
ImdbRating	Number (3,1)	By default 0
Description	Text(100)	By default Null
Metascore	Number (3,1)	By default 0
Votes	Int	By default 0
Gross	Number(10,2)	Gross amount should be greater than 10000

Table name: MoviesDirectors
Description: Used to store Movie Directors Information

Attribute	Data Type	Constraints	
MoviesId	Int	Foreign Key references Id of **Movies** table	
DirectorsId	Int	Foreign Key references Id of **Directors** table	Primary Key

Table name: MoviesStars
Description: Used to store Movie Stars Information

Attribute	Data Type	Constraints	
MoviesId	Int	Foreign Key references Id of **Movies** table	
StarsId	Int	Foreign Key references Id of **Stars** table	Primary Key

Activity # 2
Practice SQL Data Definition Language(DDL) commands

Description : 2.1 Table creation and alteration

Date : 11/08/2023

Query

1. Create the tables based on the above description.

```
SQL> create table Directors(Did int primary key, name varchar2(40) not null);

Table created.
```

```
SQL> create table stars(sid int primary key, name varchar2(40) unique, about varchar2(100));

Table created.

SQL> desc stars;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 SID                                       NOT NULL NUMBER(38)
 NAME                                               VARCHAR2(40)
 ABOUT                                              VARCHAR2(100)
```

```
SQL> create table movies(mid int primary key, title varchar2(40) not null, R_date date, Image_url varcha
r2(100), Certificate varchar2(20), Runtime Number(3,2), ImdbRating Number(3,1) Default 0, Description va
rchar2(100) Default NULL,Metascore Number(3,1) Default 0,Votes int Default(0), Gross Number(10,2) CHECK(
gross>10000));

Table created.

SQL> desc movies
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 MID                                       NOT NULL NUMBER(38)
 TITLE                                     NOT NULL VARCHAR2(40)
 R_DATE                                             DATE
 IMAGE_URL                                          VARCHAR2(100)
 CERTIFICATE                                        VARCHAR2(20)
 RUNTIME                                            NUMBER(3,2)
 IMDBRATING                                         NUMBER(3,1)
 DESCRIPTION                                        VARCHAR2(100)
 METASCORE                                          NUMBER(3,1)
 VOTES                                              NUMBER(38)
 GROSS                                              NUMBER(10,2)
```

```
SQL> create table MoviesDirectors(mid int references movies(mid), did int references directors(did),prim
ary key(mid,did));

Table created.

SQL> desc MoviesDirectors;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 MID                                       NOT NULL NUMBER(38)
 DID                                       NOT NULL NUMBER(38)
```

```
SQL> create table MoviesStars(mid int references movies(mid), sid int references stars(sid),primary key(
mid,sid));

Table created.

SQL> desc MoviesStars;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 MID                                       NOT NULL NUMBER(38)
 SID                                       NOT NULL NUMBER(38)
```

2. Add a column 'DOB' to Stars table.

```
SQL> alter table stars add dob date;

Table altered.

SQL> desc stars;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 SID                                       NOT NULL NUMBER(38)
 NAME                                               VARCHAR2(40)
 ABOUT                                              VARCHAR2(100)
 DOB                                                DATE
```

3. Drop the column 'Gross' in Movies table.

```
SQL> alter table Movies drop column gross;

Table altered.

SQL> desc Movies;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 MID                                       NOT NULL NUMBER(38)
 TITLE                                     NOT NULL VARCHAR2(40)
 R_DATE                                             DATE
 IMAGE_URL                                          VARCHAR2(100)
 CERTIFICATE                                        VARCHAR2(20)
 RUNTIME                                            NUMBER(3,2)
 IMDBRATING                                         NUMBER(3,1)
 DESCRIPTION                                        VARCHAR2(100)
 METASCORE                                          NUMBER(3,1)
 VOTES                                              NUMBER(38)
```

4. Add column 'Language' in Movies table.

```
SQL> alter table movies add language varchar2(20);

Table altered.

SQL> desc movies;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 MID                                       NOT NULL NUMBER(38)
 TITLE                                     NOT NULL VARCHAR2(40)
 RELEASEDATE                                        DATE
 IMAGE_URL                                          VARCHAR2(100)
 CERTIFICATE                                        VARCHAR2(20)
 RUNTIME                                            NUMBER(3,2)
 IMDBRATING                                         NUMBER(3,1)
 DESCRIPTION                                        VARCHAR2(100)
 METASCORE                                          NUMBER(3,1)
 VOTES                                              NUMBER(38)
 LANGUAGE                                           VARCHAR2(20)
```

5. Add column Gross Number(10,2) in Movies table

```
SQL> alter table movies add Gross number(10,2);

Table altered.

SQL> desc movies
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 MID                                       NOT NULL NUMBER(38)
 TITLE                                     NOT NULL VARCHAR2(40)
 RELEASEDATE                                        DATE
 IMAGE_URL                                          VARCHAR2(100)
 CERTIFICATE                                        VARCHAR2(20)
 RUNTIME                                            NUMBER(3,2)
 IMDBRATING                                         NUMBER(3,1)
 DESCRIPTION                                        VARCHAR2(100)
 METASCORE                                          NUMBER(3,1)
 VOTES                                              NUMBER(38)
 LANGUAGE                                           VARCHAR2(20)
 HIT                                                NUMBER(1)
 ENTRY_DATE                                         DATE
 GROSS                                              NUMBER(10,2)
```

6. Change the name of the column 'R_date' in Movies table to Releasedate. Releasedate.

```
SQL> alter table movies rename column R_date to Releasedate;

Table altered.

SQL> desc movies
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 MID                                       NOT NULL NUMBER(38)
 TITLE                                     NOT NULL VARCHAR2(40)
 RELEASEDATE                                        DATE
 IMAGE_URL                                          VARCHAR2(100)
 CERTIFICATE                                        VARCHAR2(20)
 RUNTIME                                            NUMBER(3,2)
 IMDBRATING                                         NUMBER(3,1)
 DESCRIPTION                                        VARCHAR2(100)
 METASCORE                                          NUMBER(3,1)
 VOTES                                              NUMBER(38)
 LANGUAGE                                           VARCHAR2(20)
 HIT                                                NUMBER(1)
 ENTRY_DATE                                         DATE
 GROSS                                              NUMBER(10,2)
```

7. Add a column 'Age' in Directors table as Number. Age must be 7 years or above.

```
SQL> alter table directors add age Number(10) check(age>=7);

Table altered.

SQL> desc directors
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 DID                                       NOT NULL NUMBER(38)
 NAME                                      NOT NULL VARCHAR2(30)
 AGE                                                NUMBER(10)
```

8. Add a new column 'Hit' in Movies table with datatype Number(1) and by default 0.

```
SQL> alter table Movies add hit Number(1) default 0;

Table altered.

SQL> desc movies
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 MID                                       NOT NULL NUMBER(38)
 TITLE                                     NOT NULL VARCHAR2(40)
 RELEASEDATE                                        DATE
 IMAGE_URL                                          VARCHAR2(100)
 CERTIFICATE                                        VARCHAR2(20)
 RUNTIME                                            NUMBER(3,2)
 IMDBRATING                                         NUMBER(3,1)
 DESCRIPTION                                        VARCHAR2(100)
 METASCORE                                          NUMBER(3,1)
 VOTES                                              NUMBER(38)
 LANGUAGE                                           VARCHAR2(20)
 HIT                                                NUMBER(1)
```

9. Add a new column 'Entry_date' in Movies table to record the date on which the movie details are entered in the data base.

```
SQL> alter table movies add entrydate date;

Table altered.

SQL> desc movies
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 MID                                       NOT NULL NUMBER(38)
 TITLE                                     NOT NULL VARCHAR2(40)
 RELEASEDATE                                        DATE
 IMAGE_URL                                          VARCHAR2(100)
 CERTIFICATE                                        VARCHAR2(20)
 RUNTIME                                            NUMBER(3,2)
 IMDBRATING                                         NUMBER(3,1)
 DESCRIPTION                                        VARCHAR2(100)
 METASCORE                                          NUMBER(3,1)
 VOTES                                              NUMBER(38)
 LANGUAGE                                           VARCHAR2(20)
 HIT                                                NUMBER(1)
 ENTRY_DATE                                         DATE
 GROSS                                              NUMBER(10,2)
 ENTRYDATE                                          DATE
```

10. Destroy the table MoviesStars and recreate it.

```
SQL> drop table Moviesstars;

Table dropped.

SQL> create table MoviesStars(mid int references movies(mid), sid int references stars(sid),primary key(
mid,sid));

Table created.
```

11. Change the size of the Director's name to 30.

```
SQL> alter table directors modify name varchar2(30);

Table altered.

SQL> desc directors;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 DID                                       NOT NULL NUMBER(38)
 NAME                                      NOT NULL VARCHAR2(30)
 AGE                                                NUMBER(10)
```

12. Add the following check constraints:
    - Releasedate should be less than the Entry_date in the Movies table.

```
SQL> alter table movies add check(releasedate<entry_date);

Table altered.
```

    - Language of movies should be Malayalam, English, Tamil or Hindi.

```
SQL> alter table movies add check(language in('Malayalam','English','Hindi','Tamil'));

Table altered.
```

Activity # 3
Practice SQL Data Manipulation Language (DML) commands

Description : 3.1 Row insertion, deletion and updating

Date : 11/08/2023

Query

1. Insert the appropriate data (10 rows) for the tables with respect to defined datatypes, size and constraints.

   Movies table

```
SQL> insert into movies values(101,'Inception','16-Jul-2010','http://www.movieimage.com/_Inception','PG-13',2.47,8.8,'Mind-bending thriller',74,2046235,'English',1,'01-Jan-2011',12345678.12);

1 row created.

SQL> insert into movies values(102,'Mayaanadhi','22-Dec-2022','http://www.movieimage.com/mayanadhi','U/A',2.27,8,'Romantic Drama',75,20900,'Malayalam',1,'01-Jan-2023',52345678);

1 row created.

SQL> insert into movies values(103,'Oppam','08-Sep-2016','http://www.movieimage.com/oppam','U',2.62,7.5,'Crime Thriller',70,10545,'Malayalam',0,'01-Dec-2016',1500000);

1 row created.

SQL> insert into movies values(104,'Avatar','18-Dec-2020','http://www.movieimage.com/_avatar','PG-13',2.7,7.8,'Epic sci-fi adventure',83,1181175,'English',1,'01-Sep-2021',211787.97);

1 row created.

SQL> insert into movies values(105,'Daddy Cool','14-Aug-2009','http://example.com/daddy_cool.jpg','U',3,5.8,'A quirky family comedy with a touch of mystery',5.6,4200,'Malayalam',1,'07-Sep-2009',5330000);

1 row created.

SQL> insert into movies values(106,'The Avengers','04-May-2012','http://www.movieimage.com/_avengers','PG-13',2.38,8,'Superhero ensemble',69,1353551,'English',0,'01-Sep-2013',621518.81);

1 row created.

SQL> insert into movies values(107,'Pulp Fiction','14-Oct-1994','http://www.movieimage.com/_4','R',2.57,8.9,'Quirky crime film',94,1900251,'English',1,'01-Sep-1995',5111213.93);

1 row created.

SQL> insert into movies values(109,'Lucifer','28-Mar-2019','http://example.com/lucifer.jpg','PG-13',2.5,8.5,'A political thriller with unexpected twists',8.2,32000,'Malayalam',1,'30-Mar-2019',515000);

1 row created.

SQL> insert into movies values(110,'Forrest Gump','06-Jul-1994','http://www.movieimage.com/_6','PG-13',2.37,8.8,'Heartwarming drama',82,2176822,'English',0,'01-Sep-1995',67811.18);

1 row created.

SQL> insert into movies values(108,'Master','23-Jan-2023','http://example.com/movie2.jpg','R',2,6.9,'A suspenseful thriller that takes place under the cover of darkness.',7.5,9872,'Hindi',0,'29-Aug-2023',31500.5
);

1 row created.
```

Directors Table

```
SQL> insert into directors values(115,'Priyadarshan',64);

1 row created.

SQL> insert into directors values(116,'Lijo Jose Pellissery',42);

1 row created.

SQL> insert into directors values(117,'Aashiq Abu',42);

1 row created.

SQL> insert into directors values(118,'Christopher Nolan',51);

1 row created.

SQL> insert into directors values(119,'Steven Spielberg',75);

1 row created.

SQL> insert into directors values(126,'Prithviraj Sukumaran',42);

1 row created.

SQL> insert into directors values(121,'Martin Scorsese',79);

1 row created.

SQL> insert into directors values(122,'Ridley Scott',83);

1 row created.

SQL> insert into directors values(125,'Lokesh Kanagaraj',35);

1 row created.

SQL> insert into directors values(124,'Robert Zemeckis',70);
```

Stars table

```
SQL> insert into stars values(301,'Mohanlal','Legendary Malayalam actor','21-May-1960');

1 row created.

SQL> insert into stars values(302,'Zoe Saldana','Hollywood actor','05-May-1964');

1 row created.

SQL> insert into stars values(303,'Prithviraj','Versatile actor','16-Oct-1982');

1 row created.

SQL> insert into stars values(304,'Tovino Thomas','Acclaimed actor','08-Aug-1982');

1 row created.

SQL> insert into stars values(305,'Leonardo DiCaprio','Award-winning actor','11-Nov-1974');

1 row created.

SQL> insert into stars values(306,'Robert Downey Jr.','Iron Man','04-Apr-1965');

1 row created.

SQL> insert into stars values(307,'Vijay','Popular Tamil actor','22-Jun-1974');

1 row created.

SQL> insert into stars values(308,'Emma Watson','Hermione Granger','15-Apr-1990');

1 row created.
```

```
SQL> insert into stars values(310,'Mammootty','Iconic Malayalam actor','07-Sep-1951');

1 row created.

SQL> insert into stars values(309,'Sreenath Bhasi','Malayalam actor','04-May-1985');

1 row created.
```

MoviesDirectors Table

```
SQL> insert into moviesdirectors values(101,118);

1 row created.

SQL> insert into moviesdirectors values(102,117);

1 row created.

SQL> insert into moviesdirectors values(103,115);

1 row created.

SQL> insert into moviesdirectors values(104,118);

1 row created.

SQL> insert into moviesdirectors values(105,117);

1 row created.

SQL> insert into moviesdirectors values(106,124);

1 row created.

SQL> insert into moviesdirectors values(107,122);

1 row created.

SQL> insert into moviesdirectors values(108,125);

1 row created.

SQL> insert into moviesdirectors values(109,126);

1 row created.

SQL> insert into moviesdirectors values(110,121);

1 row created.
```

Moviesstars Table

```
SQL> insert into moviesstars values(101,305);

1 row created.

SQL> insert into moviesstars values(102,304);

1 row created.

SQL> insert into moviesstars values(103,301);

1 row created.

SQL> insert into moviesstars values(104,302);

1 row created.

SQL> insert into moviesstars values(105,310);

1 row created.

SQL> insert into moviesstars values(106,306);

1 row created.

SQL> insert into moviesstars values(107,305);

1 row created.

SQL> insert into moviesstars values(108,307);

1 row created.

SQL> insert into moviesstars values(109,303);

1 row created.

SQL> insert into moviesstars values(109,301);

1 row created.
```

2. Change value of Hit to 1 where 'Votes' greater than or equal to 90.

```
SQL> update movies set hit=1 where votes>=90;

6 rows updated.
```

3. Create table **IndustryHit** with the following columns:
   - Id
   - Title
   - Releasedate
   - Language
   - Votes
   - Gross

   The data types and null characteristics for these columns should be the same as the corresponding columns in the **Movies** table described at the beginning of the lab exercise.

```
SQL> create table IndustryHit(mid int primary key, title varchar2(40) not null, ReleaseDate date, Langua
ge varchar2(20),Votes Number(38),Gross int);

Table created.

SQL> desc IndustryHit;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 MID                                       NOT NULL NUMBER(38)
 TITLE                                     NOT NULL VARCHAR2(40)
 RELEASEDATE                                        DATE
 LANGUAGE                                           VARCHAR2(20)
 VOTES                                              NUMBER(38)
 GROSS                                              NUMBER(38)

SQL>
```

4. New movies hit the box office; their data is as follows:
   - Id: 1014, 1021, 1032
   - Title: 2018: Everyone is a Hero, Oppenheimer, Maamannan
   - Releasedate: 5 May 2023, 21 July 2023, 29 June 2023
   - Language: Malayalam, English, Tamil
   - Votes: 97, 96, 95
   - Gross: 750000000, 500000000, 505000000
   - Add the new employees to the **IndustryHit** table.

```
SQL> Insert into IndustryHit values(1014,'Everyone is a hero','5-may-2023','Malayalam',97,750000000);

1 row created.

SQL> Insert into IndustryHit values(1021,'Oppenheimer','21-Jul-2023','English',96,500000000);

1 row created.

SQL> Insert into IndustryHit values(1032,'Maamannan','29-Jun-2023','Tamil',95,505000000);

1 row created.
```

5. Insert data into the new **IndustryHit** table.

```
SQL> insert into IndustryHit values(1056,'PuliMurugan','27-Oct-2023','Malayalam',98,625000000);

1 row created.

SQL> insert into IndustryHit values(1011,'Gulmohar','25-May-2023','Hindi',95,504010000);

1 row created.

SQL>  insert into IndustryHit values(1051,'Maari','09-Jan-2023','Tamil',97,514350000);

1 row created.

SQL>
```

6. Insert data into the **IndustryHit** table by copying the appropriate columns in the **Movies** table for those Movies that have Votes greater than or equal to 95.

```
SQL> Insert into IndustryHit select mid,title,releasedate,language,votes,gross from movies where votes>=
95;

2 rows created.
```

7. Movie Oppenheimer got a Metascore of 80. Make the appropriate data change.

```
SQL> update movies set metascore=80 where title='Oppenheimer';

1 row updated.
```

8. Delete all movies whose Metascore is less than 50.

```
SQL> delete from movies where metascore<50;
```

9. Movie 'Voice Of Sathyanathan' was released.
    - For 'Voice Of Sathyanathan' enter the following data:
        - Id: 1015
        - Title: Voice Of Sathyanathan
        - Releasedate: 28 July 2023
        - Image_url: https://m.media-amazon.com/imak2M_.jpg
        - Certificate: U
        - Runtime: 2.10
        - ImdbRating: 7.4
        - Description: A man's life becomes increasingly complicated after his neighbor is injured in a dispute over a fence.
        - Metascore: 60
        - Votes: 90
        - Gross: 109500000

```
SQL>  insert into movies values(1015,'Voice Of Sathyanathan','29-Jul-2023','https://m.media-amazon.com/i
mak2M_.jpg','U',2.10,7.4,'A mans life becomes increasingly complicated after his neighbor is injured in
a dispute over a fence',60,90,'01-Sep-2023','Malayalam',1,109500000);

1 row created.
```

10. Delete all rows from **IndustryHit and drop the IndustryHit table.**

```
SQL> delete from industryhit;

8 rows deleted.

SQL> drop table industryhit;

Table dropped.
```

Activity # 3
Practice SQL Data Manipulation Language (DML) commands

Description : 3.2 Retrieval of data (Simple select query and select with 'where' options (include all relational and logical operators)

Date : 17/08/2023

Query

1. List details of all movies

```
SQL> select * from movies;

       MID TITLE                                RELEASEDA IMAGE_URL
                                CERTIFICATE          RUNTIME IMDBRATING DESCRIPTION
                                METASCORE      VOTES ENTRYDATE LANGUAGE                    HIT      GROSS
---------- ---------------------------------------- --------- ---------- ---------- ---------- ----------
------------------------------ -------------------- ---------- ---------- -------------------- ---------- ----------
------------------------------------------------------------ ---------- --------- -------------------- ---------- ----------

      1015 Voice Of Sathyanathan               29-JUL-23 https://m.media-amazon.com/imak2M_.jpg
                                U                    2.1        7.4 A mans life becomes increasingly complicated after his n
eighbor is injured in a dispute over a fence       60         90 01-SEP-23 Malayalam                  1  109500000
       101 Inception                            16-JAN-20 http://www.movieimage.com/_Inception
                                PG-13                2.47        8.8 Mind-bending thriller
                                                    74         95 01-JUN-20 English                    1   12345678
       102 Mayaanadhi                           22-DEC-22 http://www.movieimage.com/mayanadhi
                                U/A                  1.27        6.4 Romantic Drama
                                                    75         94 01-JAN-23 Malayalam                  1 5234500678
       103 Oppam                                16-JUN-20 http://www.movieimage.com/oppam
                                U                    2.62        7.5 Crime Thriller
                                                    90         93 01-DEC-20 Malayalam                  1    1500000
       104 Avatar                               16-JUN-20 http://www.movieimage.com/_avatar
                                PG-13                 2.7        5.9 Epic sci-fi adventure
                                                    83         91 01-SEP-21 English                    1     211788
       105 Daddy Cool                           14-FEB-09 http://example.com/daddy_cool.jpg
                                U                      3        5.8 A quirky family comedy with a touch of mystery
                                                   5.6         89 07-SEP-09 Malayalam                  1  650030000
       106 The Avengers                         04-MAY-12 http://www.movieimage.com/_avengers
                                PG-13                2.38          8 Superhero ensemble
                                                    69         90 01-SEP-13 English                    1     621519
       107 Pulp Fiction                         14-FEB-20 http://www.movieimage.com/_4
                                R                    2.57         8.9
                                                    94         78 01-SEP-20 English                    1    5111214
       108 Master                               23-JAN-23 http://example.com/movie2.jpg
                                R                      2        6.9 A suspenseful thriller that takes place under the cover
of darkness.                                     7.5         45 29-AUG-23 Tamil                      0      31501
       109 Lucifer                              28-MAR-20 http://example.com/lucifer.jpg
                                PG-13                2.5        8.5 A political thriller with unexpected twists
                                                    8.2         34 30-MAR-20 Malayalam                  1     515000
       110 Oppenheimer                          21-FEB-23 http://example.com/openhei.jpg
                                R                    2.75        8.1
                                                    80         96 25-JUL-23 English                    1  505000000

11 rows selected.
```

2. List Title, Votes, ReleaseYear, Gross where Gross collection greater than 5000,000,00. Sequence the results in descending order by Gross.

```
SQL> select title,votes,releasedate,gross from movies where gross>=500000000 order by gross desc;

TITLE                                      VOTES RELEASEDA      GROSS
---------------------------------------- ---------- --------- ----------
Mayaanadhi                                     94 22-DEC-22 5234500678
Daddy Cool                                     89 14-FEB-09  650030000
Oppenheimer                                    96 21-FEB-23  505000000
```

3. Retrieve the titles and years of Tamil movies released in 2022.

```
SQL> select title, releasedate from movies where language='Tamil' and TO_CHAR(releasedate,'YYYY')=2022;

no rows selected
```

4. Get the titles, years, and meta scores of movies sorted in descending order of meta scores.

```
SQL> select title,releasedate,metascore from movies order by metascore desc;

TITLE                                    RELEASEDA  METASCORE
---------------------------------------- --------- ----------
Pulp Fiction                             14-FEB-20         94
Oppam                                    16-JUN-20         90
Avatar                                   16-JUN-20         83
Oppenheimer                              21-FEB-23         80
Mayaanadhi                               22-DEC-22         75
Inception                                16-JAN-20         74
The Avengers                             04-MAY-12         69
Voice Of Sathyanathan                    29-JUL-23         60
Lucifer                                  28-MAR-20        8.2
Master                                   23-JAN-23        7.5
Daddy Cool                               14-FEB-09        5.6

11 rows selected.
```

5. List titles, years, languages, dates and votes of all Malayalam and English movies released before 2022 and ImdbRating less than 7. The list should be ordered by Title.

```
SQL> select title,releasedate,language,entrydate,votes from movies where language in ('Malayalam','English') and releasedate<'01-Jan-2022' and  IMDBRATING<7 order by title;

TITLE                                    RELEASEDA LANGUAGE             ENTRYDATE      VOTES
---------------------------------------- --------- -------------------- --------- ----------
Avatar                                   16-JUN-20 English              01-SEP-21         91
Daddy Cool                               14-FEB-09 Malayalam            07-SEP-09         89
```

6. List all the movies whose title starts with ‘Oppen’. Order the result by descending order of their id.

```
SQL> select mid,title from movies where title like 'Oppen%' order by mid desc;

       MID TITLE
---------- ----------------------------------------
       110 Oppenheimer
```

7. List Hit movies released in 2022 and 2023. Order the result by ascending order of their Titles.

```
SQL> select * from movies where hit=1 and to_char(releasedate,'YYYY') in (2022,2023) order by title;

      MID TITLE                                RELEASEDA IMAGE_URL
                                                
```

9. Retrieve movies with Metascore ratings below 50 and IMDb ratings above 6.0.

```
SQL> select mid,title,metascore,imdbrating from movies where metascore <50 and imdbrating>6;

       MID TITLE                                     METASCORE IMDBRATING
---------- ---------------------------------------- ---------- ----------
       108 Master                                         7.5        6.9
       109 Lucifer                                        8.2        8.5
```

10. Retrieve movies with no description provided.

```
SQL> select * from movies where description is NULL;

       MID TITLE                          RELEASEDA IMAGE_URL                                                                CERTIFICATE              RU
NTIME IMDBRATING DESCRIPTION                                                                 METASCORE     VOTES ENTRYDATE LANGUAGE                 HIT      GROSS
---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
---
       107 Pulp Fiction               14-FEB-20 http://www.movieimage.com/_4                                                R
 2.57     8.9                                                                                94        78 01-SEP-20 English                  1    5111214

       110 Oppenheimer                21-FEB-23 http://example.com/openhei.jpg                                              R
 2.75     8.1                                                                                80        96 25-JUL-23 English                  1  505000000
```

Activity # 3
Practice SQL Data Manipulation Language (DML) commands

Description : 3.3 Functions: Numeric Data, Character Conversion and Group functions

Date : 23/08/2023

Query

1. Illustrate the different numeric functions using dual table (power, round, ceil, floor, abs, exp, greatest, least, mod, trunc, round, sign, sqrt etc.)

```
SQL> select power(5,6) ,round(80.766761,4) , ceil(80.35641), floor(59.4) , abs(6.55) , exp(25) , greatest(503,800,632), least(310,228,126), mod(25,3) , trunc(125.8150,-2) , sign(-55) , sqrt
(789) from dual;

POWER(5,6) ROUND(80.766761,4) CEIL(80.35641) FLOOR(59.4)  ABS(6.55)    EXP(25) GREATEST(503,800,632) LEAST(310,228,126)  MOD(25,3) TRUNC(125.8150,-2)  SIGN(-55)  SQRT(789)
---------- ------------------ -------------- ----------- ---------- ---------- -------------------- ------------------ ---------- ------------------- ---------- ----------
     15625            80.7668             81          59       6.55 7.2005E+10                  800                126          1                 100         -1 28.0891438
```

2. Illustrate the character functions (upper, lower, initcap, length, concat, ascii, substr, ltrim, rtrim, trim, translate, instr, chr,Lpad,Rpadetc) using the table **Movies.**

```
SQL> select upper(title) UPPER,lower(title) as LOWER,initcap(title),length(description) LENGTH,concat(mid,metascore) CONCAT,ascii(certificate) ASCII,substr(title,0,4),ltrim(title,'M') LTRIM
,rtrim(title,'r') as RTRIM,trim(title),translate(title, 'aeiou', 'AEIOU') from movies;

UPPER                                    LOWER                                    INITCAP(TITLE)                           LENGTH CONCAT
                                       ASCII SUBS LTRIM                           RTRIM                                      TRIM(TITLE)                              TRANSLATE(TITLE,'AE
IOU','AEIOU')
---------------------------------------- ---------------------------------------- ---------------------------------------- ---------- ---------------------------------------- ----------------------------------------
---------- ---- ---------------------------------------- ---------------------------------------- ---------- ---------------------------------------- ----------------------------------------
----------
VOICE OF SATHYANATHAN                      voice of sathyanathan                    Voice Of Sathyanathan                     100 101560
                                         85 Voic Voice Of Sathyanathan                Voice Of Sathyanathan                    Voice Of Sathyanathan                    VOIcE Of SAthyAnAth
An
INCEPTION                                  inception                                Inception                                  21 10174
                                         80 Ince Inception                            Inception                                Inception                                IncEptIOn
MAYAANADHI                                 mayaanadhi                               Mayaanadhi                                 14 10275
                                         85 Maya ayaanadhi                            Mayaanadhi                               Mayaanadhi                               MAyAAnAdhI
OPPAM                                      oppam                                    Oppam                                      14 10390
                                         85 Oppa Oppam                                Oppam                                    Oppam                                    OppAm
AVATAR                                     avatar                                   Avatar                                     21 10483
                                         80 Avat Avatar                               Avata                                    Avatar                                   AvAtAr
DADDY COOL                                 daddy cool                               Daddy Cool                                 46 1055.6
                                         85 Dadd Daddy Cool                           Daddy Cool                               Daddy Cool                               DAddy COOl
THE AVENGERS                               the avengers                             The Avengers                               18 10669
                                         80 The  The Avengers                         The Avengers                             The Avengers                             ThE AvEngErs
PULP FICTION                               pulp fiction                             Pulp Fiction                               10794
                                         82 Pulp Pulp Fiction                         Pulp Fiction                             Pulp Fiction                             PUlp FIctIOn
MASTER                                     master                                   Master                                     68 1087.5
                                         82 Mast aster                                Maste                                    Master                                   MAstEr
LUCIFER                                    lucifer                                  Lucifer                                    43 1098.2
                                         80 Luci Lucifer                              Lucife                                   Lucifer                                  LUcIfEr
OPPENHEIMER                                oppenheimer                              Oppenheimer                                11080
                                         82 Oppe Oppenheimer                          Oppenheime                               Oppenheimer                              OppEnhEImEr

11 rows selected.
```

3. Illustration of conversion functions- to_number, to_char(numberconversion), to_char(dateconversion)

```
SQL> SELECT TO_NUMBER('365.1'),TO_CHAR(9876.54, '9999.99'),to_char(sysdate,'DD-MM-YY') FROM dual;

TO_NUMBER('365.1') TO_CHAR( TO_CHAR(
------------------ -------- --------
             365.1  9876.54 07-09-23
```

4. Count the total no. of Movies

```
SQL> select count(*) AS COUNT from movies;

     COUNT
----------
        11
```

5. Calculate the average votes of movies.

```
SQL> select avg(votes) from movies;

AVG(VOTES)
----------
81.3636364
```

6. Determine the maximum and minimum collection of movies. Rename the output as Max_Coll and Min_Coll respectively.

```
SQL> select max(gross) Max_Coll, min(gross) Min_Coll from movies;

  MAX_COLL   MIN_COLL
---------- ----------
5234500678      31501
```

7. Count the number of movies crossed the collection 50,00,00,000.

```
SQL> select count(*) from movies where gross>500000000;

  COUNT(*)
----------
         3
```

8. Count the hit movies of 2021.

```
SQL> select count(*) from movies where hit=1 and to_char(releasedate,'YYYY')=2021;

  COUNT(*)
----------
         0
```

Activity # 3
Practice SQL Data Manipulation Language (DML) commands

Description : 3.4 Data manipulations using date functions

Date : 25/08/2023

Query

1. Provide a list of all movies which were released on June 16, 2020. Display the year and month of the released date and the Id. Sort the result by Id. Name the derived columns YEAR and MONTH.

```
SQL> select mid,title,to_char(releasedate,'YYYY') YEAR,to_char(releasedate,'MONTH') MONTH from movies where releasedate='16-Jun-2020' order by mid;

      MID TITLE                                    YEAR MONTH
--------- ---------------------------------------- ---- ---------
      103 Oppam                                    2020 JUNE
      104 Avatar                                   2020 JUNE
```

2. List the number of months between release date and entry date of each movie.

```
SQL> select mid,title,round(months_between(ENTRYDATE,RELEASEDATE))MONTHS from movies;

       MID TITLE                                     MONTHS
---------- ---------------------------------------- ----------
      1015 Voice Of Sathyanathan                             1
       101 Inception                                         5
       102 Mayaanadhi                                        0
       103 Oppam                                             6
       104 Avatar                                           15
       105 Daddy Cool                                        7
       106 The Avengers                                     16
       107 Pulp Fiction                                      7
       108 Master                                            7
       109 Lucifer                                           0
       110 Oppenheimer                                       5

11 rows selected.
```

3. List the Entry_date in the format 'DD-Month-YY'.

```
SQL> SELECT mid,title,TO_CHAR(Entrydate, 'DD-Month-YY') AS Entry_Date FROM Movies;

       MID TITLE                                     ENTRY_DATE
---------- ---------------------------------------- ---------------
      1015 Voice Of Sathyanathan                     01-September-23
       101 Inception                                 01-June     -20
       102 Mayaanadhi                                01-January  -23
       103 Oppam                                     01-December -20
       104 Avatar                                    01-September-21
       105 Daddy Cool                                07-September-09
       106 The Avengers                              01-September-13
       107 Pulp Fiction                              01-September-20
       108 Master                                    29-August   -23
       109 Lucifer                                   30-March    -20
       110 Oppenheimer                               25-July     -23

11 rows selected.
```

4. List the date, 8 days after today's date.

```
SQL> SELECT SYSDATE + 8 AS Eight_Days_After FROM DUAL;

EIGHT_DAY
---------
12-SEP-23
```

5. List all the movies which were released in the month of February.

```
SQL> select mid,title,releasedate from movies where to_char(releasedate,'mm')=02;

       MID TITLE                                    RELEASEDA
---------- ---------------------------------------- ---------
       105 Daddy Cool                               14-FEB-09
       107 Pulp Fiction                             14-FEB-20
       110 Oppenheimer                              21-FEB-23
```

6. Illustrate the different date functions using dual table (to_date, Add_months, last_day, months_between, next_day, round etc.)

```
SQL> select
  2  to_date(sysdate)TOD,add_months(sysdate,4)ADDD,last_day(sysdate)LASTD,months_between('19-mar^B2023','19-mar-2022')MONB,next_day(sysdate,'Sunday')NEXTD,round(sysdate,'YYYY')RUND from dua
l;

TOD      ADDD     LASTD          MONB NEXTD    RUND
-------- -------- -------- ---------- -------- --------
04-SEP-23 04-JAN-24 30-SEP-23         12 10-SEP-23 01-JAN-24
```

```
SQL> select
  2  to_date(sysdate)TOD,add_months(sysdate,4)ADDD,last_day(sysdate)LASTD,months_between('19-mar
l;

TOD       ADDD      LASTD           MONB NEXTD     RUND
--------- --------- --------- ---------- --------- ---------
04-SEP-23 04-JAN-24 30-SEP-23          12 10-SEP-23 01-JAN-24
```

7. Illustration of special date formats using to_char function (use of th,sp,spth)

```
SQL> select TO_CHAR(SYSDATE, 'DDth'),TO_CHAR(SYSDATE, 'DDsp'),TO_CHAR(SYSDATE, 'DDspth') from dual;

TO_C TO_CHAR(SYSD TO_CHAR(SYSDAT
---- ----------- -------------
07TH SEVEN       SEVENTH
```

8. Calculate the total gross earnings for movies released after June 16, 2020.

```
SQL> select sum(gross) from movies where releasedate>'16-jun-2020';

SUM(GROSS)
----------
5849032179

SQL>
```

Activity # 3
Practice SQL Data Manipulation Language (DML) commands

Description : 3.5 Set Operations

Date : 26/08/2023

Query

1. Create a new table IndustryHit (Id, title, genre, Certificate, Gross, Releasedate). Insert some movies from Movies table and some new movies in the new table IndustryHit.

```
SQL> create table IndustryHit(id int primary key, title varchar2(100) not null, genre varchar
2(50), certificate varchar2(10), gross int, releasedate date);

Table created.

SQL>

SQL> INSERT INTO IndustryHit (Id, Title, Certificate, Gross, Releasedate) SELECT mid, Title,
Certificate, Gross, Releasedate FROM (SELECT mid, Title, Certificate, Gross, Releasedate FROM
 Movies ORDER BY ImdbRating DESC)WHERE ROWNUM <= 5;

5 rows created.

SQL> insert into industryhit values(200,'life of pi','adventure drama','U/A',6900000000.00,'2
8-sep-2012');

1 row created.

SQL> insert into industryhit values(202,'jailer','action thriller','U/A',5660000000.00,'10-au
g-2023');

1 row created.
```

2. Retrieve the titles of all movies and industry hits which are in the action thriller genre.

```
SQL> select title from movies union select title from industryhit where genre='action thrille
r';

TITLE
--------------------------------------------------------------------------------------------------
-------
Avatar
Daddy Cool
Inception
Lucifer
Master
Mayaanadhi
Oppam
Oppenheimer
Pulp Fiction
The Avengers
Voice Of Sathyanathan

TITLE
--------------------------------------------------------------------------------------------------
-------
jailer

12 rows selected.
```

3. Retrieve the titles of all movies including industry hits.

```
SQL> select title from movies where hit=1 union select title from industryhit;

TITLE
--------------------------------------------------------------------------------------------------------
-------
Avatar
Daddy Cool
Inception
Lucifer
Mayaanadhi
Oppam
Oppenheimer
Pulp Fiction
The Avengers
Voice Of Sathyanathan
jailer

TITLE
--------------------------------------------------------------------------------------------------------
-------
life of pi

12 rows selected.
```

4. Retrieve the titles of all movies which are not industry hits.

```
SQL> SELECT Title FROM Movies MINUS SELECT Title FROM Industryhit;

TITLE
--------------------------------------------------------------------------------------------------------
-------
Avatar
Daddy Cool
Master
Mayaanadhi
Oppam
Voice Of Sathyanathan

6 rows selected.
```

Activity # 3
Practice SQL Data Manipulation Language (DML) commands

Description : 3.6 Illustration of Group By having clause

Date : 29/08/2023

Query

1. For all genres, display genre type and the sum of all Gross for each genre. Name the derived column SUM_COLL.

```
SQL> select genre,sum(gross) as sum_coll from industryhit group by genre;

GENRE                                               SUM_COLL
-------------------------------------------------- ----------
                                                    523593411
adventure drama                                    6900000000
action thriller                                    5660000000
```

2. For all genres, display the genre type and the number of titles. Name the derived column TITLE_COUNT.

```
SQL> SELECT Genre, count(Title) AS Count FROM Industryhit GROUP BY Genre;

GENRE                                                 COUNT
-------------------------------------------------- ----------
                                                           5
adventure drama                                            1
action thriller                                            1
```

3. Display the genres which have more than 3 titles.

```
SQL>  select genre,count(title) from industryhit group by genre having count(title)>3;

GENRE                                              COUNT(TITLE)
-------------------------------------------------- ------------
                                                              5
```

4. Retrieve the total number of movies released in each year, only for years with at least 5 movies.

```
SQL> select count(title) from movies group by releasedate having count(title)>5;

no rows selected
```

5. List the certificates along with the number of movies for each certificate, but only show certificates with more than 3 movies.

```
SQL> select certificate,count(title) from movies group by certificate having count(title)>3;

CERTIFICATE          COUNT(TITLE)
-------------------- ------------
PG-13                           4
```

6. Show the total gross earnings for each certificate, but only for certificates with total gross greater than $1 million.

```
SQL> select certificate,sum(gross) from movies group by certificate having sum(gross)>1000000
;

CERTIFICATE          SUM(GROSS)
-------------------- ----------
U/A                  5234500678
U                     761030000
R                     510142715
PG-13                  13693985
```

7. List the release years with the highest number of movies and the corresponding movie count, limited to the top 3 years.

```
SQL> select * FROM (select to_char(releasedate,'YYYY') YEAR,count(*) Count FROM Movies GROUP BY to_char(releasedate,'YYYY') ORDER BY Count DESC) where ROWNUM<=3;

YEAR      COUNT
---- ----------
2020          5
2023          3
2009          1
```

Activity # 3
Practice SQL Data Manipulation Language (DML) commands

Description : 3.7 Sub queries

Date : 01/09/2023

Query

1. Retrieve the titles and runtime of movies with the highest Metascore.

```
SQL> SELECT Title, Runtime FROM Movies WHERE Metascore = (SELECT MAX(Metascore) FROM Movies);


TITLE                                       RUNTIME
----------------------------------------- ----------
Pulp Fiction                                  2.57
```

2. List the titles of movies with a Gross amount greater than the average Gross amount of all movies.

```
SQL> select title from movies where gross >(select avg(gross) from movies);

TITLE
----------------------------------------
Mayaanadhi
Daddy Cool
```

3. Retrieve the titles and descriptions of movies with a Metascore lower than the average Metascore.

```
SQL> select title,description from movies where metascore <(select avg(metascore) from movies
);

TITLE                                     DESCRIPTION
----------------------------------------- --------------------------------------------------
--------------------------------------------------
Daddy Cool                                A quirky family comedy with a touch of mystery
Master                                    A suspenseful thriller that takes place under the co
ver of darkness.
Lucifer                                   A political thriller with unexpected twists

SQL>
```

4. List the movie titles and their IMDb ratings for movies released in the year with the highest average IMDb rating.

```
SQL> SELECT Title, ImdbRating, to_char(releasedate,'YYYY') Release_Year FROM Movies WHERE to_char(releasedate,'YYYY') = (select year from (SELECT to_char(releasedate,'YYYY') year,round(AVG(
har(releasedate,'YYYY') order by Avg(imdbrating) desc)where rownum=1);

TITLE                                     IMDBRATING RELE
----------------------------------------- ---------- ----
The Avengers                                       8 2012
```

5. Retrieve the movie titles and their IMDb ratings for movies that have a Metascore greater than twice their IMDb rating.

```
SQL> SELECT title, imdbRating FROM movies WHERE metascore > ALL (SELECT
  2  POWER(imdbRating, 2) FROM movies);

TITLE                                    IMDBRATING
---------------------------------------- ----------
Oppenheimer                                     8.1
Avatar                                          5.9
Oppam                                           7.5
Pulp Fiction                                    8.9
```

6. Find the title and gross amount of the top 3 highest-grossing movies.

```
SQL> SELECT Title, Gross FROM (select * from movies order by gross desc ) where
  2  rownum <=3;

TITLE                                        GROSS
---------------------------------------- ----------
Mayaanadhi                               5234500678
Daddy Cool                                650030000
Oppenheimer                               505000000
```

7. Calculate the total number of votes received by movies released in the year 2022.

```
SQL> SELECT SUM(Votes) AS Total_Votes FROM (SELECT * FROM MOVIES
  2  WHERE EXTRACT(Year from Releasedate)=2022);

TOTAL_VOTES
-----------
         94
```

8. List the titles and certificate ratings of movies that have an IMDb rating below the average IMDb rating.

```
SQL> SELECT Title, Certificate FROM movies WHERE ImdbRating < (SELECT
  2  AVG(ImdbRating) FROM Movies);

TITLE                                    CERTIFICATE
---------------------------------------- --------------------
Voice Of Sathyanathan                    U
Mayaanadhi                               U/A
Avatar                                   PG-13
Daddy Cool                               U
Master                                   R
```

Activity # 3
Practice SQL Data Manipulation Language (DML) commands

Description : 3.8 Views

Date : 14/09/2023

Query

1. Create a view called MovieDetails that combines information from the Movies, Directors, and Stars tables to display movie titles, directors' names, and the names of stars who acted in those movies.

```
SQL> CREATE VIEW MovieDetails as select title, d.name
DirectorName, s.name StarName from movies m,directors d,stars
s,moviesstars ms, moviesdirectors md where m.mid=ms.mid and
d.did=md.did and s.sid=ms.sid and m.mid=md.mid;

View created.

SQL> SELECT * FROM MovieDetails;

TITLE                                          DIRECTORNAME
STARNAME
---------------------------------------- ------------------
------------ ----------------------------------------
Inception                                   Christopher Nolan
Leonardo DiCaprio
Mayaanadhi                                        Aashiq Abu
Tovino Thomas
Oppam                                          Priyadarshan
Mohanlal
Avatar                                      Christopher Nolan
Zoe Saldana
Daddy Cool                                        Aashiq Abu
Mammootty
The Avengers                                 Robert Zemeckis
Robert Downey Jr.
Pulp Fiction                                    Ridley Scott
Leonardo DiCaprio
Master                                      Lokesh Kanagaraj
Vijay
Lucifer                                   Prithviraj Sukumaran
Mohanlal
Lucifer                                   Prithviraj Sukumaran
Prithviraj

10 rows selected.
```

2. Create a view called HighlyRatedMovies that displays movies with IMDb ratings greater than 8.0, including their titles and ratings.

```
SQL> Create View HighlyratedMovies as select
title,imdbrating from movies where imdbrating>8;

View created.

SQL> SELECT * FROM HighlyRatedMovies;

TITLE                                    IMDBRATING
---------------------------------------- ----------
Inception                                       8.8
Pulp Fiction                                    8.9
Lucifer                                         8.5
Oppenheimer                                     8.1
```

3. Create a view called DirectorMovies that provides a list of directors along with the number of movies they have directed.

```
SQL>  Create View DirectorMovies as select d.name
Director,count(*) No_of_Movies from moviesdirectors
md,directors d where d.did=md.did group by d.name;

View created.

SQL> select * from DirectorMovies;

DIRECTOR                  NO_OF_MOVIES
------------------------- ------------
Prithviraj Sukumaran                 1
Lokesh Kanagaraj                     1
Martin Scorsese                      1
Robert Zemeckis                      1
Christopher Nolan                    2
Aashiq Abu                           2
Ridley Scott                         1
Priyadarshan                         1

8 rows selected.
```

4. Create a view called StarMovies that displays stars' names and the titles of movies they have acted in.

```
SQL> Create View StarsMovies as select s.name
Stars,m.title Movie from stars s,movies m,moviesstars
ms where m.mid=ms.mid and s.sid=ms.sid;

View created.
```

```
SQL> Select * from StarsMovies;

STARS                                     MOVIE
----------------------------------------  -------------
---------------------------
Leonardo DiCaprio                         Inception
Tovino Thomas                             Mayaanadhi
Mohanlal                                  Oppam
Zoe Saldana                               Avatar
Mammootty                                 Daddy Cool
Robert Downey Jr.                         The Avengers
Leonardo DiCaprio                         Pulp Fiction
Vijay                                     Master
Prithviraj                                Lucifer
Mohanlal                                  Lucifer

10 rows selected.
```

5. Create a view called LongestMovies that lists the titles of movies with the longest runtimes (duration).

```
SQL> Create View LongestMovies as select title,runtime
from movies where runtime=(select max(runtime) from
movies);

View created.

SQL> Select * from LongestMovies;

TITLE                                       RUNTIME
----------------------------------------- ----------
Daddy Cool                                         3
```

6. Create a view called LanguageDistribution that shows the distribution of movies based on the languages they were released in, including the count of movies for each language.

```
SQL> Create view LanguageDistribution as select
language,count(*) COUNT from movies group by language;

View created.

SQL> Select * from LanguageDistribution;

LANGUAGE                    COUNT
-------------------- ----------
Malayalam                    5
Tamil                        1
English                      5
```

7. Create a view called GrossEarnings that displays movies with their titles and gross earnings, sorted by earnings in descending order.

```
SQL> Create View GrossEarnings as select title,gross
from movies order by gross desc;

View created.

SQL> Select * from GrossEarnings;

TITLE                                       GROSS
---------------------------------------- ----------
Mayaanadhi                              5234500678
Daddy Cool                               650030000
Oppenheimer                              505000000
Voice Of Sathyanathan                    109500000
Inception                                 12345678
Pulp Fiction                               5111214
Oppam                                      1500000
The Avengers                                621519
Lucifer                                     515000
Avatar                                      211788
Master                                       31501

11 rows selected.
```

8. Create a view called IndustryHitMovies that shows the titles and release dates of industry hit movies.

```
SQL> Create View IndustryHitMovies as select title
Movie,Releasedate from movies where hit=1;

View created.

SQL> Select * from IndustryHitMovies;

MOVIE                                    RELEASEDA
---------------------------------------- ---------
Voice Of Sathyanathan                    29-JUL-23
Inception                                16-JAN-20
Mayaanadhi                               22-DEC-22
Oppam                                    16-JUN-20
Avatar                                   16-JUN-20
Daddy Cool                               14-FEB-09
The Avengers                             04-MAY-12
Pulp Fiction                             14-FEB-20
Lucifer                                  28-MAR-20
Oppenheimer                              21-FEB-23

10 rows selected.
```

9. Create a view called MovieVotes that displays movies along with their titles and the number of votes they have received.

```
SQL> Create View MovieVotes as select title movie,votes
from movies;

View created.

SQL> Select * from MovieVotes;

MOVIE                                        VOTES
---------------------------------------- ----------
Voice Of Sathyanathan                            90
Inception                                        95
Mayaanadhi                                       94
Oppam                                            93
Avatar                                           91
Daddy Cool                                       89
The Avengers                                     90
Pulp Fiction                                     78
Master                                           45
Lucifer                                          34
Oppenheimer                                      96

11 rows selected.
```

10. Create a view called CertifiedMovies that lists movies with their titles and certificates (e.g., U/A, U).

```
SQL> Create View CertifiedMovies as select title
Movie,Certificate from movies;

View created.

SQL> Select * from CertifiedMovies;

MOVIE                                    CERTIFICATE
---------------------------------------- -------------
-------
Voice Of Sathyanathan                    U
Inception                                PG-13
Mayaanadhi                               U/A
Oppam                                    U
Avatar                                   PG-13
Daddy Cool                               U
The Avengers                             PG-13
Pulp Fiction                             R
Master                                   R
Lucifer                                  PG-13
Oppenheimer                              R
11 rows selected.
```

Activity # 4
Practice PL/SQL

Description : 4.1 Introduction to PL/SQL

Date : 16-09-2023

Query

1. Write a PL/SQL code block to calculate the area of a circle for a value of radius varying from 3 to 7. Store the radius and corresponding values of calculated area in an empty table named Areas, consisting of two columns Radius and Area.

```
SQL> CREATE TABLE Circle(Radius NUMBER, Area Number);

Table created.

SQL> edit E:\MCA_2333\pg1.sql

 DECLARE
  i  number;
  a Number;
 BEGIN
   for i
   in 3..7
   loop
     a:=3.14*i*i;
     insert into circle values(i,a);
   end loop;
 END;


SQL> @ E:\MCA_2333\pg1.sql
12 /

PL/SQL procedure successfully completed.

SQL> select * from circle;

    RADIUS       AREA
---------- ----------
         3      28.26
         4      50.24
         5       78.5
         6     113.04
         7     153.86
```

2. Write a PL/SQL block of code for inverting a number accepted from the console.

```
SQL> edit E:\MCA_2333\pg2.sql

DECLARE
  n number;
  r number;
BEGIN
  n:=&n;
  r:=0;
  while n>0 loop
    r:=(r*10)+mod(n,10);
    n:=floor(n/10);
  end loop;
  dbms_output.put_line('Inverted Number : '||154);
END;

SQL> @ E:\MCA_2333\pg2.sql
13  /
Enter value for n: 451
old   5:   n:=&n;
new   5:   n:=451;
Inverted Number : 154

PL/SQL procedure successfully completed.
```

3. Write a PL/SQL code block that will accept an account number from the user and debit an amount of Rs.2000 from the account if the account has a minimum balance of 500 after the amount is debited. The process is fired on the Accounts table.

```
SQL> create table account(accno int,name
Varchar2(15),balaance number(10,2));
 Table created.

 SQL> insert into account values(3004,'Mariya',200000);

 1 row created.

 SQL> insert into account values(3001,'Megha',500600);

 1 row created.

 SQL> insert into account values(3001,'Merin',1000);

 1 row created.


SQL> edit E:\MCA_2333\pg3.sql

declare
   acno int;
```

```
   bal number(10,2);
 begin
   acno:=&acno;
   select balaance into bal from account where
accno=acno;
   if bal-2000>500 then
     bal:=bal-2000;
     update account set balaance=bal where accno=acno;
   else
     dbms_output.put_line('Transaction Not posiible');
   end if;
end;

SQL> @ E:\MCA_2333\pg3.sql
14  /
Enter value for acno: 3004
old   5:    acno:=&acno;
new   5:    acno:=3004;

PL/SQL procedure successfully completed.

SQL> select * from account;

     ACCNO NAME            BALAANCE
---------- --------------- ----------
      3004 Mariya            198000
      3001 Megha             500600
      3001 Merin               1000
```

4. Write a PL/SQL block of code that updates the salaries of Maria Jacob and Albert by Rs. 2000/- and Rs.2500/- respectively. Then check to see that the total salary does not exceed 75000. If the total salary is greater than 75000, then undo the updates made to salaries of both. (Use savepoint, rollback and commit).

```
SQL> create table employ(empno int,name varchar2(20),salary
number(10,2));

Table created.

SQL> insert into employ values(101,'Maria',20000);

1 row created.

SQL> insert into employ values(102,'Albert',15000);

1 row created.

SQL> insert into employ values(103,'Megha',20000);

1 row created.
```

```
SQL> edit E:\MCA_2333\pg4.sql

declare
   s number(10,2);
begin
   savepoint s1;
   update employ set salary=salary+2000 where empno=101
or empno=102;
   select sum(salary) into s from employ;
   if s >75000 then
     rollback to s1;
     dbms_output.put_line('Rollbacked');
   else
     commit;
   end if;
end;

SQL> @ E:\MCA_2333\pg4.sql
14  /

PL/SQL procedure successfully completed.
```

Activity # 4
Practice PL/SQL

Description : 4.2.1 Illustration of Implicit Cursors

Date : 21-09-2023

Query

1. Write a PL/SQL block to accept an employee number and update the salary of that employee to raise the salary by 0.15. Display appropriate message based on the existence of the record in the employee table.

```
SQL> edit E:\MCA_2333\pg5.sql

declare
begin
  update employee set salary=salary+(salary*0.15) where
empno=&empno;
  if sql%found then
   dbms_output.put_line(sql%rowcount||' records updated');
  end if;
end;

SQL> select empno,salary from employee where empno='E0340';

EMPNO      SALARY
---------- ----------
E0340           900

SQL> @ E:\MCA_2333\pg5.sql
 10  /
Enter value for empno: 'E0340'
old   4:   update employee set salary=salary+(salary*0.15)
where empno=&empno;
new   4:   update employee set salary=salary+(salary*0.15)
where empno='E0340';
1 records updated

PL/SQL procedure successfully completed.

SQL> select empno,salary from employee where empno='E0340';

EMPNO      SALARY
---------- ----------
E0340           1035
```

2. The HRD manager decides to raise the salary of employees working as 'analyst' by 0.15. Write a cursor to update the salary of the employees. Display the no. of employee records that has been modified.

```
SQL> edit E:\MCA_2333\pg6.sql

declare
begin
  update employee set salary=salary+(salary*0.15)
where job='ANALYST';
  if sql%found then
   dbms_output.put_line(sql%rowcount||' records
updated');
  end if;
end;

SQL> @ E:\MCA_2333\pg6.sql
10  /

2 records updated
PL/SQL procedure successfully completed.
```

Activity # 4
Practice PL/SQL

Description : 4.2.2 Illustration of Explicit Cursors

Date : 21-09-2023

Query

1. Write an explicit cursor to display the name,department, salary of the first 5 employees getting the highest salary.

```
SQL> edit E:\MCA_2333\pg15.sql

declare
     CURSOR empcur IS select * from (select
empname,deptname,salary from employee e, department d
where e.deptno=d.deptno order by salary desc ) where
rownum<=5;
   empv empcur%rowtype;
begin
  dbms_output.put_line('Employee Details');
  open empcur;
  fetch empcur into empv;
  while empcur%found loop
     dbms_output.put_line('Empnname   : '||empv.empname);
     dbms_output.put_line('Department : '||empv.deptname);
     dbms_output.put_line('Salary     : '||empv.salary);
     dbms_output.put_line('---------------------------');
     fetch empcur into empv;
  end loop;
  close empcur;
  end;

SQL> @ E:\MCA_2333\pg15.sql
 17  /
Employee Details
Empnname   : ARNOLD LEONARD AMON
Department : COMPUTER SERVICE DIVISION
Salary     : 152750
---------------------------
Empnname   : DONA ANICE SIBY
Department : COMPUTER SERVICE DIVISION
Salary     : 46500
---------------------------
Empnname   : PHILIP VINCENT
Department : PLANNING
Salary     : 41250
---------------------------
Empnname   : ALFRIN LUIZ
Department : SUPPORT SERVICES
```

```
Salary     : 40175
----------------------------
Empnname   : SHILVY K K
Department : INFORMATION CENTER
Salary     : 38250
----------------------------

PL/SQL procedure successfully completed.
```

2. The HRD manager decides to raise the salary of employees working as 'analyst' by 0.15. Whenever any such raise is given to the employees, a record for the same is maintained in the emp_raise table. It includes the employee number, the date when the raise was given and actual raise. Write a PL/SQL block to update the salary of the employees and insert a record in the emp_raise table.

Emp_raise(empcode, raisedate,raise_amt)

```
SQL> create table emp_raise (empcode varchar(10),
raisedate date, raise_amt number(10,2));

Table created.

SQL> edit E:\MCA_2333\pg16.sql

- code
declare
  CURSOR empcur IS select * from employee where
job='ANALYST';
  empv empcur%rowtype;
  sal number(10,2);
begin
  open empcur;
  fetch empcur into empv;
  while empcur%found loop
     sal:=(empv.salary*0.15)+empv.salary;
     insert into emp_raise
values(empv.empno,sysdate,sal);
     fetch empcur into empv;
  end loop;
  dbms_output.put_line(empcur%rowcount||'  rows
updated');
  close empcur;
  end;
SQL> @ E:\MCA_2333\pg16.sql

 16  /

2  rows updated
```

```
PL/SQL procedure successfully completed.

SQL> select * from emp_raise;

EMPCODE   RAISEDATE  RAISE_AMT
---------- --------- ----------
E0130     23-SEP-23      27370
E0140     23-SEP-23      32683
```

Activity # 4
Practice PL/SQL

Description : 4.3 Illustration of Procedures

Date : 15-10-2023

Query

1. Write a PL/SQL block which makes use of a stored procedure Proj_emp (emp_name varchar2(50)) which finds all the details of the projects involved by the given employee.

```
edit C:\Users\cclab29\OneDrive\mca2333\p22.sql
create or replace procedure Proj_emp(ename in varchar)
as
 CURSOR projcur IS select
ep.empno,empname,ep.projno,projname from emp_pro
ep,project p,employee e where e.empname=ename and
ep.empno=e.empno AND p.projno=ep.projno;
  empv projcur%rowtype;

begin
 dbms_output.put_line('Project Details');
  open projcur;
  fetch projcur into empv;
  while projcur%found loop
    dbms_output.put_line('Empnname    :
'||empv.empname);
    dbms_output.put_line('ProjNo    : '||empv.projno);
    dbms_output.put_line('projname  :
'||empv.projname);
    dbms_output.put_line('----------------------------
');
    fetch projcur into empv;
  end loop;
  close projcur;
end;

SQL> @ C:\Users\cclab29\OneDrive\mca2333\p22.sql
 20  /

Procedure created.
SQL> execute PROJ_EMP('RIYA TONEY');
Project Details
Empnname        : RIYA TONEY
ProjNo    : AD3112
projname  : PERSONNEL PROGRAMMING
----------------------------
```

```
Empnname        : RIYA TONEY
ProjNo     : IF2000
projname   : USER EDUCATION
----------------------------

PL/SQL procedure successfully completed.
```

2. Write a procedure to check whether a string is a palindrome. Call the procedure to list all the palindrome names in the employee table.

```
edit C:\Users\cclab29\OneDrive\mca2333\p23.sql

create or replace procedure palindrome(n in varchar)
as
s varchar2(30);
begin
  select reverse(n) into s from dual;
  if n=s then
     dbms_output.put_line(n);
  end if;
end;

SQL> @ C:\Users\cclab29\OneDrive\mca2333\p23.sql
 10  /

Procedure created.
SQL> edit
C:\Users\cclab29\OneDrive\mca2333\p23_main.sql
declare
 cursor enamcur is select empname from employee;
 name employee.empname%type;
begin
  open enamcur;
  fetch enamcur into name;
  while enamcur%found loop
     palindrome(name);
     fetch enamcur into name;
   end loop;
   close enamcur;
end;

SQL> set serveroutput on;
SQL> @ C:\Users\cclab29\OneDrive\mca2333\p23_main.sql
 13  /
ABCCBA

PL/SQL procedure successfully completed.
```

3. Write a PL/SQL block which retrieve all the employee into a cursor and display the details of all assigned projects for each employee using a stored procedure Proj_emp (emp_name varchar2(50)).

```
SQL> edit C:\Users\cclab29\OneDrive\mca2333\proce.sql

create or replace procedure Proj_emp(ename in char)
as
 CURSOR projcur IS select
ep.empno,empname,ep.projno,projname from emp_proj
ep,project p,employeee e where e.empname=ename and
ep.empno=e.empno AND p.projno=ep.projno;
  empv projcur%rowtype;

begin
 dbms_output.put_line('Project Details');
  open projcur;
  fetch projcur into empv;
  while projcur%found loop
        dbms_output.put_line('EmpnNO    :
'||empv.empno);
     dbms_output.put_line('Empnname     :
'||empv.empname);
     dbms_output.put_line('ProjNo    :
'||empv.projno);
     dbms_output.put_line('projname  :
'||empv.projname);
     dbms_output.put_line('-------------------------
-');
     fetch projcur into empv;
  end loop;
  close projcur;

end;
SQL> @ C:\Users\cclab29\OneDrive\mca2333\proce.sql

20  /

Procedure created.

SQL> edit C:\Users\cclab29\OneDrive\mca2333\pro_m.sql
declare
  CURSOR empcur IS select empname from employeee;
  enov employeee.empname%type;
begin
  open empcur;
  fetch empcur into enov;
  while empcur%found loop
     Proj_emp(enov);
```

```
     fetch empcur into enov;
  end loop;
  close empcur;
  end;
SQL> @ C:\Users\cclab29\OneDrive\mca2333\pro_m.sql
13  /
Project Details
Empnname        : ARNOLD LEONARD AMON
ProjNo    : AD3100
projname  : ADMIN SERVICES
---------------------------
Empnname        : ARNOLD LEONARD AMON
ProjNo    : MA2100
projname  : WELD LINE AUTOMATION
---------------------------
Project Details
Empnname        : PHILIP VINCENT
ProjNo    : AD3100
projname  : ADMIN SERVICES
---------------------------
Empnname        : PHILIP VINCENT
ProjNo    : MA2111
projname  : W L PROGRAM DESIGN
---------------------------
Empnname        : PHILIP VINCENT
ProjNo    : PL2100
projname  : WELD LINE PLANNING
---------------------------
Project Details
Empnname        : SHILVY K K
ProjNo    : IF1000
projname  : QUERY SERVICES
---------------------------
Empnname        : SHILVY K K
ProjNo    : IF2000
projname  : USER EDUCATION
---------------------------
Project Details
Empnname        : ALFRIN LUIZ
ProjNo    : OP1000
projname  : OPERATION SUPPORT
---------------------------
Empnname        : ALFRIN LUIZ
ProjNo    : OP2000
projname  : GEN SYSTEM SERVICES
---------------------------
Project Details
Empnname        : OSHINA ANTONY
ProjNo    : AD3100
projname  : ADMIN SERVICES
---------------------------
```

```
Empnname        : OSHINA ANTONY
ProjNo    : MA2110
projname  : W L PROGRAMMING
----------------------------
Project Details
Empnname        : BINCY PAUL
ProjNo    : AD3110
projname  : GENERAL AD SYSTEMS
----------------------------
Project Details
Empnname        : ANAMIKA PAUL
ProjNo    : AD3110
projname  : GENERAL AD SYSTEMS
----------------------------
Empnname        : ANAMIKA PAUL
ProjNo    : OP1010
projname  : OPERATION
----------------------------
Project Details
Empnname        : ANEESH DENNY
ProjNo    : AD3110
projname  : GENERAL AD SYSTEMS
----------------------------
Empnname        : ANEESH DENNY
ProjNo    : OP2010
projname  : SYSTEMS SUPPORT
----------------------------
Project Details
Empnname        : DONA ANICE SIBY
ProjNo    : AD3113
projname  : ACCOUNT PROGRAMMING
----------------------------
Project Details
Empnname        : JUNAID K V
ProjNo    : MA2111
projname  : W L PROGRAM DESIGN
----------------------------
Project Details
Empnname        : CHRISTEENA THOMAS
ProjNo    : AD3113
projname  : ACCOUNT PROGRAMMING
----------------------------
Empnname        : CHRISTEENA THOMAS
ProjNo    : MA2111
projname  : W L PROGRAM DESIGN
----------------------------
Project Details
Empnname        : JEFFIN DOMINIC
ProjNo    : AD3100
projname  : ADMIN SERVICES
----------------------------
```

```
Empnname        : JEFFIN DOMINIC
ProjNo    : AD3112
projname  : PERSONNEL PROGRAMMING
---------------------------
Project Details
Empnname        : JEWEL BIJOY
ProjNo    : MA2112
projname  : W L ROBOT DESIGN
---------------------------
Empnname        : JEWEL BIJOY
ProjNo    : OP2000
projname  : GEN SYSTEM SERVICES
---------------------------
Empnname        : JEWEL BIJOY
ProjNo    : OP2010
projname  : SYSTEMS SUPPORT
---------------------------
Project Details
Empnname        : MELLOW REEBA THOMAS
ProjNo    : OP2000
projname  : GEN SYSTEM SERVICES
---------------------------
Project Details
Empnname        : VISHAK VIJAYAKUMAR
ProjNo    : PL2100
projname  : WELD LINE PLANNING
---------------------------
Project Details
Empnname        : CORRINE ELIZABETH RODRIGUES
ProjNo    : MA2112
projname  : W L ROBOT DESIGN
---------------------------
Project Details
Empnname        : MERLIN M.D
ProjNo    : MA2112
projname  : W L ROBOT DESIGN
---------------------------
Project Details
Empnname        : VISHALAKSHI V PRABHU
ProjNo    : AD3111
projname  : PAYROLL PROGRAMMING
---------------------------
Project Details
Empnname        : ANGEL PAUL
ProjNo    : AD3111
projname  : PAYROLL PROGRAMMING
---------------------------
Project Details
Empnname        : RIYA TONEY
ProjNo    : AD3112
projname  : PERSONNEL PROGRAMMING
```

```
----------------------------
Empnname        : RIYA TONEY
ProjNo    : IF2000
projname  : USER EDUCATION
----------------------------
Project Details
Empnname        : PRIYA TOMY
ProjNo    : IF1000
projname  : QUERY SERVICES
----------------------------
Empnname        : PRIYA TOMY
ProjNo    : IF2000
projname  : USER EDUCATION
----------------------------
Project Details
Empnname        : ARYAMOL ASOKAN
ProjNo    : AD3113
projname  : ACCOUNT PROGRAMMING
----------------------------
Empnname        : ARYAMOL ASOKAN
ProjNo    : IF2000
projname  : USER EDUCATION
----------------------------
Project Details
Project Details
Empnname        : JIMMY THOMSON
ProjNo    : OP1010
projname  : OPERATION
----------------------------
Project Details
Empnname        : ALAN PAYYAPPILLY
ProjNo    : OP2011
projname  : SCP SYSTEM SUPPORT
----------------------------
Project Details
Empnname        : BEN PETER MATHEW
ProjNo    : OP1000
projname  : OPERATION SUPPORT
----------------------------
Project Details
Empnname        : KRISHNANUNNI S
ProjNo    : OP2011
projname  : SCP SYSTEM SUPPORT
----------------------------
Project Details
Empnname        : AHALYA V A
ProjNo    : MA2100
projname  : WELD LINE AUTOMATION
----------------------------
Empnname        : AHALYA V A
ProjNo    : OP2012
```

```
projname  : APPLICATIONS SUPPORT
----------------------------
Project Details
Empnname        : ANJALI NAIR
ProjNo    : MA2110
projname  : W L PROGRAMMING
```

Activity # 4
Practice PL/SQL

Description : 4.4 Illustration of functions

Date : 18-10-2023

Query

1. Write a function to find the reverse of EmpNo in Employee table and display the EmpNo and Reversed(Emp No) of the first 5 employees using an SQL Query.

```
SQL> edit C:\Users\cclab29\OneDrive\mca2333\func1.sql


CREATE OR REPLACE FUNCTION Reversed(empno in varchar2)
return varchar2
IS
  rev varchar2(20);
BEGIN
  select reverse(empno) into rev from dual;
  return rev;
END;

SQL> @ C:\Users\cclab29\OneDrive\mca2333\func.sql

9 /

Function created.

SQL> select eno,ename,Reversed(eno) from employee where
rownum<=5;


ENO        ENAME                REVERSED(ENO)
---------- -------------------- ----------------------
E0010      MEGHA                0100E

E0011      MARIYA               1100E

E0012      MERIN                2100E

E0013      SARA                 3100E

E0014      JOHN                 4100E
```

2. Write a function that would check for the existence of an employee in the employee table given an EmpNo. If existing employee, check whether he is the manager of any department and display messages accordingly.

```
SQL> edit C:\Users\cclab29\OneDrive\mca2333\func2.sql

CREATE OR REPLACE FUNCTION CHECK_emp(empno in varchar2)
return varchar2
AS
  j varchar2(20);
BEGIN
   SELECT job into j from emp_tab where eno=empno;
   if sql%notfound then
      return('EMPLOYEE NOT FOUND');
   elsif j='MANAGER' then
      return('MANAGER');
   else
      return('NOT MANAGER');
   end if;
END;

SQL> @ C:\Users\cclab29\OneDrive\mca2333\func2.sql

15  /

Function created.

SQL> select check_emp('E0010') from dual;

CHECK_EMP('E0010')

------------------------

NOT MANAGER

SQL> insert into emp_tab values ('E0020','SYED',9000,
'MANAGER');

1 row created.

SQL> select check_emp('E0020') from dual;

CHECK_EMP('E0020')

-------------------------

MANAGER
```

Activity # 4
Practice PL/SQL

Description : 4.5 Illustration of Triggers

Date : 20-10-2023

Query

1. Consider the table Employee. Write PL/SQL statements to create a trigger when fired checks the operation performed on a table and based on the operation, a variable is assigned the value 'update' or 'delete'. Previous values of the modified record of the table Employee are stored into the appropriate variables declared and inserted to the audit table AuditEmployee.

```
SQL>     Create     table     AuditEmployee(eno
varchar2(5),ename varchar2(20), esal number(10,2),
job varchar2(20), Audit_action varchar2(20));

Table created.

SQL> edit C:\Users\cclab29\OneDrive\mca2333\p23.sql

Create or Replace TRIGGER trigaudit BEFORE delete or
update on Emp_tab
for each row
DECLARE
  audit_action varchar2(20);
BEGIN
  if deleting then
      audit_action:='DELETE';
  elsif updating then
      audit_action:='UPDATE';
  end if;
  insert into AuditEmployee
values(:old.eno,:old.ename,:old.esal,:old.job,audit_ac
tion);
END;

SQL> @ C:\Users\cclab29\OneDrive\mca2333\p23.sql

13  /

Trigger created.
```

```
SQL> update emp_tab set esal=10000 where
eno='E0013';

1 row updated.

SQL> delete from emp_tab where eno='E0013';

1 row deleted.

SQL> select * from AuditEmployee;

ENO   ENAME                ESAL  JOB           AUDIT_ACTION
----- ------------------ ---------- ------------- --------
-------
E0013 SARA                 17000 PROFESSOR          DELETE
E0013 SARA                 17000 PROFESSOR          UPDATE
```

2. Write PL/SQL statements to create a trigger which generates an error messages if the salary is below or beyond the valid range 0-5000 on the employee table. The triggering events are update and insert.

```
SQL> edit C:\Users\cclab29\OneDrive\mca2333\p24.sql

create or replace trigger Trig2 after INSERT OR UPDATE
ON  emp_tab
for each row
begin
if  (:new.esal not between 0 and 5000 )  then
   RAISE_APPLICATION_ERROR(-20500,'SALARY RANGE NOT
BETWEEN 0 AND 5000');
END IF;
END;

SQL> @ C:\Users\cclab29\OneDrive\mca2333\pg27.sql
  8  /
Trigger created.

SQL> update emp_tab set esal=6000 where eno='E0010';
update emp_tab set esal=6000 where eno='E0010'
     *
ERROR at line 1:
ORA-20500: SALARY RANGE NOT BETWEEN 0 AND 5000
ORA-06512: at "MCA.TRIG2", line 3
ORA-04088: error during execution of trigger
'MCA.TRIG2'
```

3. Write PL/SQL statements to create a trigger that limits the DML actions to the Employee table to weekdays from 8.30am to 6.30pm. If a user tries to insert/update/delete a row in the Employee table, a warning message will be prompted.

```
SQL> edit C:\Users\cclab29\OneDrive\mca2333\p25.sql

create or replace trigger mytrig1 BEFORE DELETE OR
INSERT OR UPDATE ON  emp_tab
begin
if  (to_char(sysdate,'day') in ('sun','mon')) or
(to_char(sysdate ,'hh:mi') not between  '08:30' and
'18:30')  then
RAISE_APPLICATION_ERROR(-20500,'TABLE IS SECURED');
END IF;
END;

SQL> @ C:\Users\cclab29\OneDrive\mca2333\p25.sql

  7  /

Trigger created.

SQL> insert into emp_tab
values('E0013','SARA',17000,'PROFESSOR');
insert into emp_tab
values('E0013','SARA',17000,'PROFESSOR')
            *
ERROR at line 1:
ORA-20500: TABLE IS SECURED
ORA-06512: at "SYSTEM.UP_TRIG", line 3
ORA-04088: error during execution of trigger
'SYSTEM.UP_TRIG'
```